

UNIT-4

BASICS OF R:

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Contents

- R Basics
- data types and objects
- control structures
- data frame
- Feature Engineering
- scaling, Label Encoding and One Hot Encoding
- Reductioncomplete material for this topics of data science subject

1. Introduction to R

R is an open-source programming language mainly used for:

- Data Analysis
- Statistical Computing
- Machine Learning
- Data Visualization
- Data Science Research

Features of R

- Open Source
- Cross Platform
- Large Community Support
- Excellent Visualization Libraries
- Powerful Statistical Functions

2. R Basics

2.1 Installing R

Software Required

- RStudio
- R

Simple R Program

```
print("Welcome to R Programming")
```

Output

```
[1] "Welcome to R Programming"
```

3. Data Types in R

Data types define the type of data stored in variables.

Data Type	Description	Example
Numeric	Decimal values	10.5
Integer	Whole numbers	5L
Character	Text/String	"R Language"
Logical	TRUE/FALSE	TRUE
Complex	Complex numbers	3+2i

Example of Data Types

x <- 10.5

y <- 5L

z <- "Data Science"

a <- TRUE

b <- 3 + 2i

print(x)

print(y)

print(z)

print(a)

print(b)

4. Objects in R

Objects are variables used to store data.

Creating Objects

```
name <- "Nirav"
```

```
age <- 35
```

```
print(name)
```

```
print(age)
```

Types of Objects in R

Object Type	Description
Vector	Collection of same type elements
Matrix	Two-dimensional data
Array	Multi-dimensional data
List	Different data types together
Data Frame	Table format data

5. Vectors in R

Vector stores similar type values.

Example

```
marks <- c(70, 80, 90, 85)
```

```
print(marks)
```

Operations on Vector

```
sum(marks)
```

```
mean(marks)
```

```
max(marks)
```

```
min(marks)
```

6. Matrix in R

Matrix contains rows and columns.

Example

```
mat <- matrix(c(1,2,3,4,5,6), nrow=2, ncol=3)
```

```
print(mat)
```

7. Lists in R

Lists can store multiple data types.

Example

```
student <- list(  
  name = "Rahul",  
  age = 21,  
  marks = 85  
)
```

8. Control Structures in R

Control structures control the flow of execution.

8.1 if Statement

```
x <- 10
```

```
if(x > 5){  
  print("x is greater than 5")  
}
```

8.2 if-else Statement

```
marks <- 40  
  
if(marks >= 50){  
  print("Pass")  
} else {  
  print("Fail")  
}
```

next Example

```
for(i in 1:5){  
  if(i == 3){  
    next  
  }  
  print(i)  
}
```

8.3 else-if Ladder

```
marks <- 75

if(marks >= 90){
  print("Grade A")
} else if(marks >= 70){
  print("Grade B")
} else {
  print("Grade C")
}
```

8.4 for Loop

```
for(i in 1:5){
  print(i)
}
```

8.5 while Loop

```
x <- 1

while(x <= 5){
  print(x)
  x <- x + 1
}
```

8.6 repeat Loop

```
x <- 1

repeat{
  print(x)
  x <- x + 1
  if(x > 5){
    break
  }
}
```

9. Functions in R

Functions are reusable blocks of code.

Example

```
add <- function(a, b){  
  return(a + b)  
}
```

```
result <- add(5, 3)
```

```
print(result)
```

10. Data Frame in R

A data frame is a table-like structure.

Creating Data Frame

```
student <- data.frame(  
  Name = c("Amit", "Rahul", "Priya"),  
  Age = c(20, 21, 19),  
  Marks = c(85, 90, 88)  
)  
print(student)
```

Accessing Data Frame

```
student$Name  
student$Marks
```

Summary of Data Frame

```
summary(student)
```

Adding New Column

```
student$Grade <- c("A", "A+", "A")  
  
print(student)
```

Removing Column

```
student$Grade <- NULL
```

11. Feature Engineering

Feature Engineering is the process of creating useful input variables for Machine Learning.

Importance

- Improves model accuracy
- Reduces complexity
- Handles missing values
- Converts categorical data

Types of Feature Engineering

Technique	Purpose
Scaling	Normalize values
Label Encoding	Convert categories into numbers
One Hot Encoding	Convert categories into binary columns
Feature Reduction	Reduce unnecessary features

12. Scaling

Scaling standardizes numerical data.

Types of Scaling

1.Min-Max Scaling

2.Standard Scaling

12.1 Min-Max Scaling

Formula:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Example

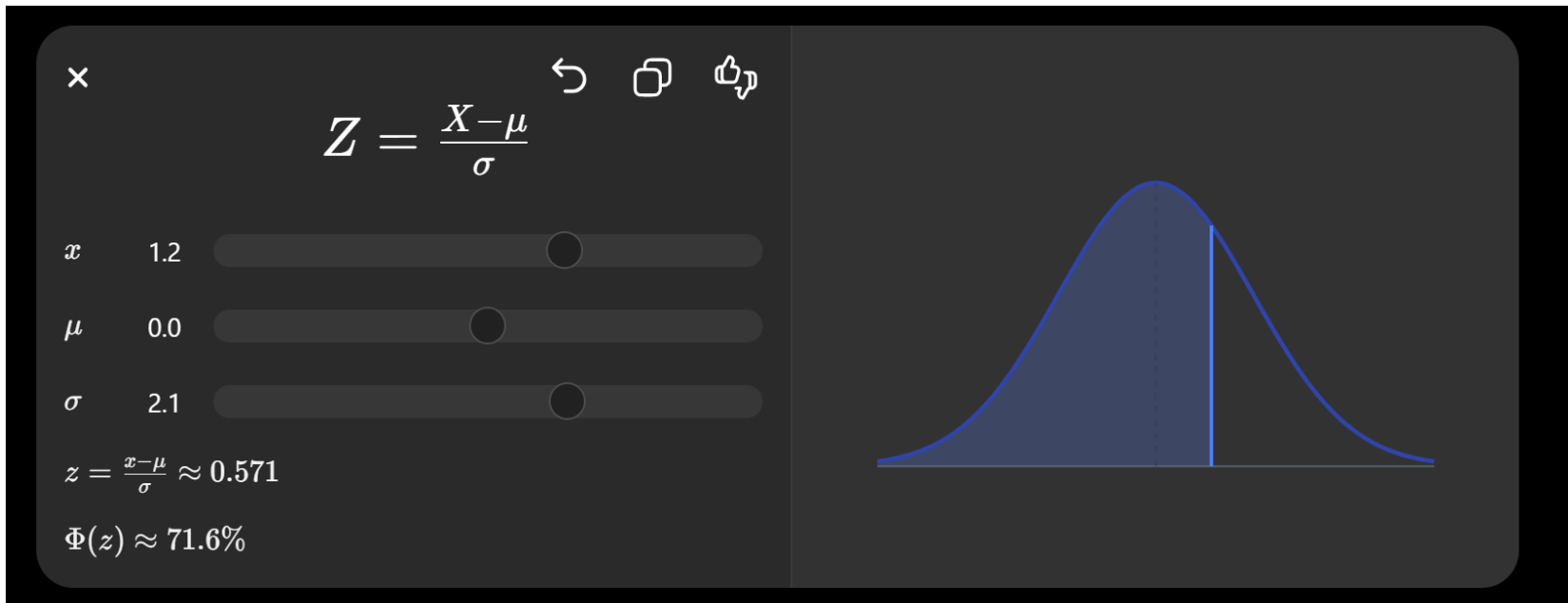
```
x <- c(10,20,30,40,50)
```

```
scaled_x <- (x - min(x)) / (max(x) - min(x))
```

```
print(scaled_x)
```


12.2 Standard Scaling

Formula:



Example

```
x <- c(10, 20, 30, 40, 50)
```

```
scaled_x <- scale(x)
```

```
print(scaled_x)
```

13. Label Encoding

Label Encoding converts categories into numeric labels.

Example

```
colors <- c("Red","Blue","Green","Red")
```

```
encoded <- as.numeric(as.factor(colors))
```

```
print(encoded)
```

Output

```
[1] 3 1 2 3
```

14. One Hot Encoding

One Hot Encoding creates binary columns for categories.

Example

```
data <- data.frame(Color = c("Red","Blue","Green"))
```

```
model.matrix(~Color-1, data)
```

Output

ColorBlue	ColorGreen	ColorRed
0	0	1
1	0	0
0	1	0

15. Feature Reduction

Feature Reduction reduces unnecessary features.

Advantages

- Faster training
- Less memory usage
- Better accuracy
- Reduces overfitting

Methods of Feature Reduction

Method	Description
Feature Selection	Select important features
PCA	Principal Component Analysis
Correlation Removal	Remove highly related features

15.1 Correlation Example

```
cor(mtcars)
```

15.2 Principal Component Analysis (PCA)

PCA reduces dimensions of data.

Example

```
data <- iris[,1:4]
```

```
pca <- prcomp(data, scale=TRUE)
```

```
summary(pca)
```

16. Important Built-in Datasets in R

Dataset	Purpose
iris	Flower classification
mtcars	Car data
airquality	Air pollution analysis
Titanic	Survival prediction

17. Data Import in R

Read CSV File

```
data <- read.csv("student.csv")  
print(data)
```

18. Data Export in R

Write CSV File

```
write.csv(data, "output.csv")
```

19. Basic Data Visualization in R

Bar Plot

```
x <- c(10,20,30)  
barplot(x)
```

Pie Chart

```
x <- c(20,30,40)  
pie(x)
```

Histogram

```
hist(mtcars$mpg)
```

20. Applications of R in Data Science

- Machine Learning
- Statistical Analysis
- Data Visualization
- Predictive Analytics
- Research
- Business Intelligence

21. Advantages of R

- Free and Open Source
- Strong Visualization
- Large Package Support
- Good for Statistical Analysis

22. Limitations of R

- Memory Consumption
- Slower for Large Data
- Complex Syntax for Beginners

Long Questions

- 1.Explain control structures in R.
- 2.Describe data frame with example.
- 3.Explain feature engineering techniques.
- 4.Explain PCA in detail.
- 5.Write a program for one hot encoding in R.

24. Conclusion

R is one of the most powerful tools in Data Science and Machine Learning. Understanding data types, objects, control structures, data frames, and feature engineering techniques like scaling, label encoding, one hot encoding, and reduction is essential for building accurate predictive models and performing efficient data analysis.

Reference Books for Basics of R

Sr. No.	Book Name	Author	Suitable For
1	R for Data Science	Hadley Wickham & Garrett Golemund	Beginners & Data Science
2	Hands-On Programming with R	Garrett Golemund	Beginners
3	R in Action	Robert Kabacoff	Intermediate Learners
4	The Art of R Programming	Norman Matloff	Advanced Concepts
5	R for Everyone	Jared P. Lander	Students & Academics
6	Practical Data Science with R	Nina Zumel & John Mount	Practical Applications
7	An Introduction to R	W.N. Venables & D.M. Smith	Official Beginner Guide

Official Reference Links

[R Project for Statistical Computing](#)

Free Online Book – R for Data Science

[R for Data Science Book](#)

Hands-On Programming with R

[Hands-On Programming with R](#)

CRAN Repository (R Packages)

[CRAN Repository](#)

RStudio / Posit Official Website

[Posit \(RStudio\) Official Website](#)

Free Introduction to R PDF

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